

An Interpretable Deep Learning Framework with Attention-Enhanced VGG19 for Pneumonia Detection

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Abstract—Pneumonia is a severe respiratory disease and a leading cause of death in the world especially in children and the aged. The Chest X-ray (CXR) imaging has been extensively used on diagnosis because it is cost-effective and it is also a universal tool, but manual interpretation is time consuming and it has an inter-observer variability. Despite achievement of promising results with deep learning methods in automated pneumonia detection, a variety of currently used models have a high level of computational complexity and low interpretability, which limits their application in the clinical environment. To overcome these issues, this paper suggests an interpretable deep learning model, which is an attention-enhanced VGG19 architecture, to detect pneumonia in pediatric chest X-rays. The VGG19 backbone is pretrained with a lightweight spatial attention mechanism on which the attention is focused on clinically significant regions of the lung and the irrelevant background information is removed. Moreover, regularization and controlled fine-tuning strategy is used in order to enhance generalization in small and unbalanced datasets. There is competitive classification performance on an accessible pediatric CXR dataset, as proved by its experimental results. Grad-CAM visualizations also help to increase the level of transparency as they use model predictions that can be explained as clinically meaningful.

Index Terms—Deep learning; Convolutional neural networks (CNN); Attention mechanism; VGG19; Transfer learning; Explainable artificial intelligence (XAI); Computer-aided diagnosis (CAD); Medical image analysis.

I. INTRODUCTION

A. Motivation

The condition of pneumonia remains a significant social health burden, an important cause of death, especially in children and the elderly demographic groups of the world. The most common diagnostic modality is the Chest X-ray (CXR) as it is cost-effective, accessible and the radiation is less harmful. Nevertheless, proper interpretation of CXR images needs well trained radiologists and may be time-consuming

and this fact renders large-scale screening and early diagnosis challenging particularly in resource-limited healthcare facilities. The recent developments in deep learning have shown promising outcomes in automated pneumonia detection using CXR images. In spite of these developments, most of the current models are based on complicated architectures with using high computational power, which prevents their application in practical clinical settings. Furthermore, the fact that many deep learning models cannot be interpreted by clinicians diminishes clinical trust because clinical professionals need reliable, clear, and explainable predictions to aid the process of diagnostic decision-making. These issues prompt the creation of effective and understandable deep learning models that could be helpful to clinicians by offering precise, dependable, and clarifiable detection of pneumonia on the basis of chest X-ray photographs.

B. Research Gap

Despite the high number of the deep learning methods that have been offered to pneumonia detection, there are a number of constraints that are not addressed. Most of the best performing models are based on deep or ensemble models that add to the complexity of computation and cannot be used in real-time in the clinical setting. Transformer-based models can be applied effectively, but can be expensive using large-scale datasets and large compute resources and are less effective with small or unbalanced medical datasets. Moreover, the majority of developed strategies are more oriented on the accuracy of the classification and neglect the interpretability of the model, which is paramount to the clinical implementation. Studies that combine lightweight attention with pretrained convolutional networks are limited and do not provide clinically meaningful explanations along with better performance.

Furthermore, poor management of the imbalance in the classes and overfitting still influence the generalization of models. In order to fill in these gaps, the interpretable, computationally efficient deep learning framework which balances accuracy, robustness and explainability is required. The paper intends to address this gap by proposing an attention-enhanced VGG19 architecture with fine-tuning and explainability mechanisms under control to achieve good pneumonia detection in Chest X-ray images.

C. Contributions

In order to cancel the failings of the unbalanced information, low interpretability and inefficient attention experienced in the earlier pneumonia detection frameworks, the paper establishes the best Attention-VGG19 model, which enhances the functionality and concentration elucidation. The principal findings of this study are as follows:

- **Regulated Fine-Tuning Strategy:** Rather than fine-tuning all the network layers, the last four convolutional layers of VGG19 are fine-tuned using a small learning rate. L2 and dropout (0.5) regularization to prevent overfitting as well as stabilize the learning at the expense of a robust generalization to the various subsets of the data.
- **Class-Balanced Training:** As the number of normal images and pneumonia images is not evenly balanced, the weight of the classes is computed dynamically to balance the loss. This will prevent the bias of the model to the majority, and better sensitivity of pneumonia detection.
- **Comprehensive Preprocessing and Augmentation:** The weight of the classes is also dynamically obtained because the number of normal and pneumonia images is uneven, so as to balance the loss function. This will help avoid the biasing of the model to the majority and sensitivity to identify pneumonia.
- **Comparative Performance Evaluation:** The proposed Attention-VGG19 is directly compared to the traditional CNN, ResNet-50, Vision Transformer (ViT). The proposed model has the strongest generalization and the best accuracy (94.1) through experiment.
- **Explainability and Clinical Interpretability:** Grad-CAM images are generated to reveal the area/s of the lung that influence the model predictions. The heatmaps support the fact that the attention module is efficient in focusing the model to pathology-specific areas to facilitate trust and transparency adoption to embrace clinical adoption.

II. RELATED WORK

Pneumonia is a severe breathing disorder and a major cause of death in the world especially in children and adults who are old. The chest X-ray (CXR) imaging is the most commonly used mode of diagnosis because it is inexpensive, broadly available, and has low radiation. Nevertheless, CXR images can be manually analyzed, which is time-consuming and may be subject to inter-observer variability, and the creation of automated computer-aided diagnosis (CAD) systems that

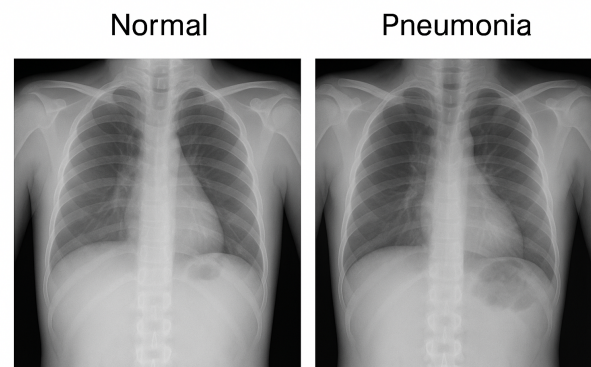


Fig. 1: Normal pediatric chest X-rays (left), Pneumonia (right).

utilize machine learning and deep learning methods has been driven by the need to automate the analysis of CXR images shown in fig.1.

Initial studies on automated pneumonia detection based on machine learning used conventional methods of machine learning with handcrafted feature extraction. It was found that deep learning models could automatically learn to extract discriminative features using CXR images with the advent of convolutional neural networks (CNNs). The performance of CheXNet, a deep CNN architecture by Rajpurkar et al., became a well-established benchmark in deep learning-based CAD systems, able to detect pneumonia on a par with a radiologist [1]. Markedly, early CNN-based methods used to be constrained by the size of the dataset and class imbalances [2]. Following research on pretrained CNNs architecture including VGG-based pneumonia classifier. Sharma and Guleria was able to show that neural networks coupled with VGG-16 was capable of achieving good performance on classification of various CXR datasets [3]. Although effective, VGG-based models are computationally intense and are likely to overfit, especially when trained on small medical data.

Ensuring robustness and generalization, the proposal of ensemble learning with a combination of multiple CNN architectures was suggested. Kundu et al. demonstrated that ensembles of GoogLeNet, ResNet, and DenseNet are better in providing stability in classification than single-model methods [4]. Nevertheless, the higher computation cost makes them not applicable in real time clinical applications.

More recent studies examined frameworks based on object-detection to pneumonia localization. Anchor-free deep learning detector that was proposed by Wu et al. in reference to RSNA dataset generated increased flexibility and localization capability [5]. However, models of this kind can be associated with intricate architectural design and hyperparameter optimization.

CNNs have also been replaced with transformer-based architectures. Singh et al. showed that Vision Transformer (ViT)-based models have the ability to extract global contextual information of CXR images [6]. Transformer-based methods

are however usually expensive in terms of data and computing, restricting their current use in small and low balance medical datasets.

A number of survey papers also point to the ongoing issues related to the automated pneumonia detection, such as class imbalance, dataset bias, and limited interpretability. As per reviews of Alapat et al. [7] and Siddiqi and Javaid [8], there is a necessity of lightweight, interpretable, and clinically robust deep learning models. Such observations encourage the creation of attention-enhanced architectures, which are balanced in terms of accuracy, efficiency, and explainability. As shown by the authors, large-scale, multinational datasets are effective in detecting COVID-19 pneumonia based on deep learning and emphasize the significance of the diversity of the data to achieve effective clinical performance (Harmon et al. [9]). A framework proposing the use of transfer learning with a name Deep-COVID is suggested by Minaee et al. [10] and demonstrated a high accuracy on the chest X-ray images with a limited amount of training data. Apostolopoulos and Mpesiana [11] demonstrated that transfer learning with convolutional neural networks can effectively detect COVID-19 pneumonia from chest X-ray images even with limited training data, highlighting the suitability of pretrained CNN models for rapid and reliable medical image diagnosis. The deep residual networks (ResNet) were proposed by He et al. [12] and are hugely popular as backbone architectures in the medical image analysis. Tang et al. [13] also integrated both deep learning and machine learning approaches to the classification of pneumonia images with a better performance of the model by incorporating model fusion techniques.

III. METHODOLOGY

The proposed system will identify pneumonia through pediatric chest X-rays based on the capability of a deep learning system that integrates transfer learning, attention mechanisms, and balanced training. The whole process comprises of four core processes, such as, information preprocessing, information improvement, framework and training plan.

- **Dataset Description:** The dataset used in the current study consists of 5,856 labeled images. They include 4,273 pneumonia (bacterial or viral) and 1,583 normal. The dataset is further divided into three subsets to ensure effective model learning, parameter tuning, and performance evaluation. Each of the pictures has been downsampled to 224 x 224 pixels to match the input size of VGG19, as well as downsampled to pixel intensities of [0,1].
- **Preprocessing:** The image was subjected to a series of preprocessing operations in order to enhance the overall quality of the image and to ensure a consistent quality of the images throughout the dataset before training the model. Firstly, the 3x 3 kernel of a median filter was applied to remove salt and pepper noise and other minor artifacts in images that ensured smoother texture of the lung textures. To make the visibility even better, the processes of the histogram equalization and brightness

normalization have been conducted, providing the contrast and displaying finer structural aspects of the lung areas. In addition, the coercion of the marginal text markers, labels, and other unfit contents that can distort the model when extracting the features, was done by the implementation of, cropping and masking techniques. Finally, all images were made to zero mean and unit variance, so that there was equal strength everywhere and less difficult and consistent convergence in training.

- **Data Augmentation:** Training was done using the ImageDataGenerator module that would facilitate the data augmentation in order to avoid overfitting and enhance the ability of the model in generalizing to various imaging conditions. The augmentation process had limited options to the input pictures like, randomly rotating the images to a maximum of 15 degree, horizontal flips, zooming to a range of 0.9 to 1.1 and increasing or decreasing the brightness value by 0.8 to 1.2. The changes presented above succeeded in raising the variety of the training data, which enabled the model to remain resilient to the alterations in the position of the patient, light, and the imaging apparatus. The model later on had the ability to detect pneumonia related characteristics with more reliability even in alternate radiographic settings.
- **Proposed Model:** Attention-VGG19: The proposed network is based on the VGG19 architecture that is originally pre-trained on ImageNet and provides a solid base of generic visual features. The initial and middle convolutional layers were kept fixed but the last four layers were trained to detect pneumonia-specific patterns. The final convolutional stage was followed by an attention block that aids the model in being diagnostically focused on the parts of the lungs that are important. The attention module comprises two 1×1 convolutional layers and a sigmoid activation which produces an attention mask:

$$A(x) = \sigma(W_2 * \text{ReLU}(W_1 * x)) \quad (1)$$

Classification Layers: Spatial dimensions are reduced with (GAP) layer:

- A 512 neuron Dense layer, ReLU-activated, L2 regularized.
- A Dropout layer (0.5) in order to avoid overfitting.
- A last Dense layer (1 neuron, Sigmoid activation) to categorize the input as to whether it is a case of Normal or Pneumonia.

A. Training Strategy

The training strategy was designed to ensure stable convergence. The network was optimized using the Adam optimizer with a learning rate of 1×10^{-5} , providing a good balance between convergence speed and stability during fine-tuning. Binary cross-entropy loss was employed, which is suitable for binary classification tasks such as pneumonia detection. To address class imbalance, class weights were computed using the *compute class weight* function from the scikit-learn library, ensuring that pneumonia and normal samples

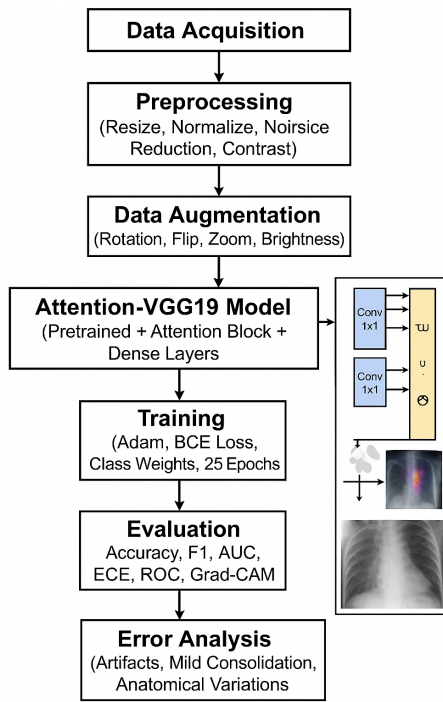


Fig. 2: Model training and evaluation pipeline.

contributed equally during training. The model was trained in a GPU-enabled Google Colab environment to facilitate efficient parallel computation. Early stopping and validation monitoring were employed to prevent overfitting and to select the best-performing model. This training strategy improved model generalization while maintaining stable learning performance across epochs shown in fig.2.

IV. EVALUATION METRICS

Accuracy is used to measure the global accuracy of the model by determining the percentage of cases that are correctly classified (with and without pneumonia) of all the cases considered. Precision concentrates on the consistency of the positive prediction of the model, that is, it points to the number of the cases that were diagnosed as pneumonia yet were in fact pneumonia. Recall (or sensitivity) measures the model performance in terms of identifying the pneumonia cases, and it indicates the number of the true pneumonia cases that were correctly identified. Additionally, the Expected Calibration Error (ECE) was also added to measure the extent of the correspondence between the predicted probabilities and the real outcomes such that the confidence of the model in its predictions are comparable to the results that are produced in reality.

V. EXPERIMENTAL RESULTS

A. Quantitative Performance

Table I presents the comparative results of the suggested Attention-VGG19 model versus the baseline architectures.

B. Learning Curves

Grad-CAM visualization shows that the suggested model is able to localize the areas of pneumonia in the lungs, which proves that the attention module is targeting the areas that are clinically significant. The ROC Curve was analyzed using a different program, as shown below. According to the analysis of the ROC curve, it is clear that Attention-VGG19 has the largest AUC of 0.97, indicating the best capacity to differentiate pneumonia and normal cases. Grad-CAM explainability: It is an approach in which an existing model is used to determine parts of an image that indicate particular features. According to the outcomes of Grad-CAM images, the suggested model is productive in identifying the location of pneumonia on the lung X-rays, which demonstrates that the attention module is successful in the highlighting of the clinically significant locations.

C. Model Comparisons

The suggested Attention-VGG19 model is more precise and superior to the baseline CNN, ResNet-50, and CPNet models, having an accuracy of 94.1 percent and an F1-score of 93.8 percent, which is approximately 3 percent and 2.8 percent higher in accuracy and F1-score respectively compared to CPNet. The important causes of this improvement are as follows:

- **Attention Mechanism:** The model allows paying attention to the diagnostically significant areas of the lungs and mitigating interference with extraneous structures or artifacts.
- **Fine-Tuning:** This allows the final convolutional layers to specialize on pneumonia and this specialization is further enhanced.
- **Regularization and Class Balancing:** The reduction of overfitting by L2 regularization and dropout and the reduction of the imbalance in the dataset by class weighting.

The benefit of ResNet-50 is that, it has residual connections that boost gradient flow and feature reuse, but may fail to detect fine-grained local textures which are relevant in CXRs of pediatric patients. CPNet despite the fact that it can support dense layers and dropout lacks explicit spatial attention and might be described with low interpretability.

D. Calibration and Reliability

In this section, the calibration and reliability of the system will be demonstrated. The question of calibration is extremely significant when it comes to clinical use of AI, since too confident or too cautious predictions may result in erroneous decision-making. The Attention-VGG19 model has the lowest Expected Calibration Error (ECE) and this implies that the probability which is predicted is as close to the actual probability as possible. On the other hand, the values of ECE of ResNet-50 and ViT are higher, meaning that their estimates of the confidence cannot be trusted in clinical practice.

TABLE I: Comparison of the performance of the proposed Attention-VGG19 and baseline models.

Description/Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC
CNN (Baseline)	87.2	86.8	87.0	86.9	0.89
ResNet-50	89.3	89.6	89.0	89.2	0.91
CPNet (VGG19 variant)	91.0	91.6	90.9	90.6	0.94
Attention-VGG19 (Proposed)	94.1	94.3	93.4	93.8	0.97

E. Error Taxonomy

According to the misclassification analysis, it is possible to enumerate three primary causes of error:

- Type A - Artifacts Tubes, lines or text markers will occasionally lead to a false positive.
- Type B - Anatomical Variations: The model is confused by chest irregularities such as unusual heart or lung shape sometimes.

Grad-CAM images show that Attention-VGG19 model tends to focus on the corresponding parts of the body, which means that the rest of the errors may be explained by the fact of the insufficient data instead of the current design of the model. This indicates that there may be additional misclassifications that may be reduced through the features in the preprocessing, artifact removal, multi-scale feature extraction, and diversity of the dataset.

F. Clinical Implications

The excellent accuracy, good calibration, and interpretability of the Attention-VGG19 model render the model applicable to three primary uses: triage assistance, second-opinion assistance, and large-scale screening of children. To be successfully deployed, it needs:

- Internet connectivity to hospital PACS systems to allow a seamless workflow.
- An interface that is easy to use with prediction and Grad-CAM heatmap available to radiologists.
- Constant evaluation and feedback mechanisms to retrain the model on new cases, which makes sure that performance is maintained in the long-run.

Altogether, Attention-VGG19 represents a combination of the latest feature extraction and clinically explainable predictions, which points out its possible use in real-world pediatric pneumonia detection.

G. Federated Learning

The collaboration of training between hospitals that do not exchange patient data can improve generalization without breaching privacy.

H. Multi-Class Classification

Going beyond binary classification to make a distinction between bacterial, viral, and COVID-19 pneumonia cases.

I. Edge Optimization

Modelling quantization and pruning can facilitate the use of portable radiography machines in low-resource medical environments.

J. Temporal Learning

The longitudinal diagnosis and monitoring might be enhanced with the use of temporal medical histories or sequential imaging data.

VI. DEPLOYMENT CONSIDERATIONS

To apply an AI model in practice in a real clinical environment, one will have to overcome the problems of system integration, privacy, and usability:

- **PACS Integration:** The model can be incorporated into hospital Picture Archiving and Communication Systems (PACS) using DICOM interfaces to enable automatic analysis of new chest X-rays to enable a seamless workflow.
- **Edge vs. Cloud Inference:** Local (edge) deployment can provide faster inference and enhance the privacy of data of resource-constrained hospitals, whereas the cloud-based deployment can enable the scalability, which is centralized.
- **Compliance and Security:** Local (edge) deployment can provide faster inference and enhance the privacy of data of resource-constrained hospitals, whereas the cloud-based deployment can enable the scalability, which is centralized.
- **Radiologist-Friendly Dashboard:** It is a user-friendly tool that helps a clinician to make a decision by giving predictions, confidence ratings, and Grad-CAM heatmaps.

Reproducibility is essentially exercised on an open scientific research, and there is a series of steps that were undertaken to render the experiments reasonably reproducible. All the experiments were set up and documented with their software (TensorFlow 2.17.1, Keras 2.17.0, scikit-learn 1.3.0), hardware (NVIDIA GPU). To make the training logs, model definitions and preprocessing scripts reproducible, the training logs, model definitions and preprocessing scripts have been openly stored in open repository. All the experiments had the consistency of the data pipeline, centralized policy of the data division, normalization, and augmentation. Parameters and learning rates, batch sizes and hyperparameters of the optimization were also automatically tracked to allow them to be traced. The combination of these practices assists other researchers in having a great framework to be in a position to replicate this work, confirm and elaborate with a lot of confidence.

VII. CONCLUSION

The article is a superior and innovative method of diagnosing pneumonia using the X-rays of the chest by adding an attention mechanism to the VGG19 model. This extra attention, as well as regularization and balanced learning, does not only aid in the enhancing of the results of the model, but also enable the model to make more intuitive decisions. Grad-CAM visualizations also can further be used to understand what the model is focused on, which is critical to clinical trust. The system will be further enhanced in the future with hybrid attention transformers, federated learning, and edge optimization, which will enable the system to facilitate healthcare and telemedicine in various regions.

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